Sic Bo Big-Small Simulator

1. Mathematical Foundations

This application will simulate the *Sic Bo* environment, a three fair 6-sided dice game, for studying probability distributions and some classical inferential statistics. This application will specifically study on the *Big* and *Small* outcomes.

The Sample Space (Ω): The experiment utilizes three independent, fair six-sided dice. The total combinations are $6^3 = 216$ equiprobable combination of outcomes.

Outcome Definition

- ***Small:*** The sum of the dice ($S = \sum d_i$) satisfies $4 \leq S \leq 10$, excluding triples.
- ***Big:*** The sum of the dice ($S = \sum d_i$) satisfies $11 \leq S \leq 17$, excluding triples.

The House Advantage: In standard *Sic Bo* rules, *Small* (sums of 4–10) and *Big* (sums of 11–17) both **lose if a triple appears**.

Theoretical Probability

For each of *Small* and *Big*, the total number of favorable outcomes are

$$P(Small) = P(Big) = \frac{105}{216} \approx 0.4861$$

Outcome	Sum of Dice	Triple Excluded?	Total Combinations	Net Favorable (Wins)
Small	4	No	3	3
	5	No	6	6
	6	Yes (2-2-2)	10	9
	7	No	15	15
	8	No	21	21
	9	Yes (3-3-3)	25	24
	10	No	27	27
Total Small			107	105
Big	11	No	27	27
	12	Yes (4-4-4)	25	24
	13	No	21	21
	14	No	15	15
	15	Yes (5-5-5)	10	9
	16	No	6	6
	17	No	3	3
Total Big			107	105

2. Computational Framework & Shiny Architecture

The application uses the *shinydashboard* framework to create the application environment.

UI Component Design

- **Sidebar Control Plane:** Houses *numericInput* for p, *sliderInput* for n, and the "RUN SIMULATION" trigger.
- **Dashboard Body:** Displays real-time metrics through *valueBox* and visualizes data via *ggplot2* passed to *plotOutput* for visual analytics.

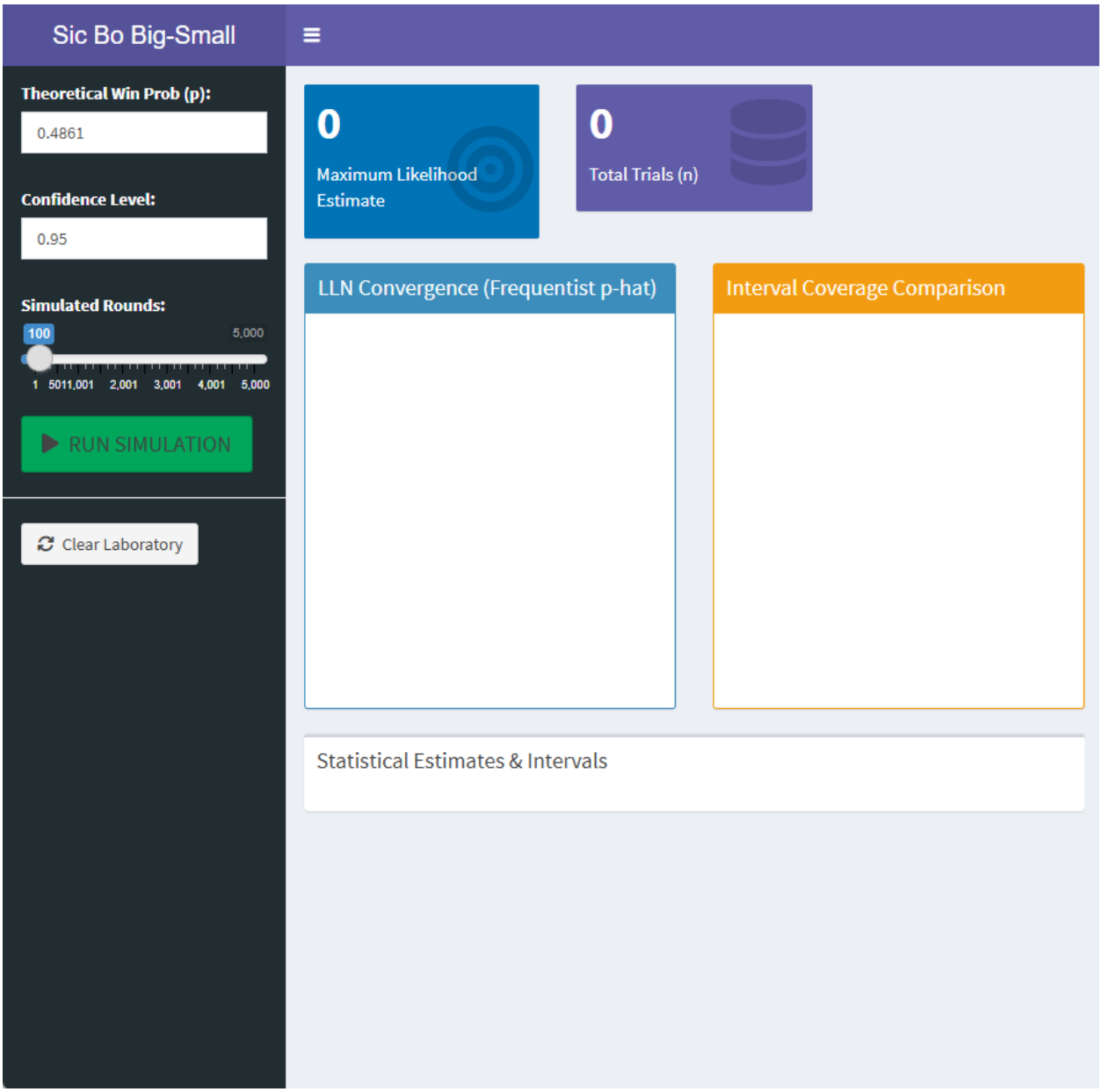

Figure 1. The R Shiny dashboard in its initial state. The sidebar displays the default theoretical parameters, while the dashboard body features the reactive containers for the Maximum Likelihood Estimate, Law of Large Number Convergence Trace, and Interval Coverage Comparisons.

Server-Side Reactive Logic

The engine leverages *reactiveValues* to store cumulative data. This allows for persistent state management, where every new simulation run appends to previous data.

- **Generation:** Trials are generated via *rbinom(n, 1, p)*, where 1 represents a "Success" (win) and 0 represents "Failure".
    - *observeEvent(input$play):* The event handler executing *rbinom(n, 1, p)*.
- **Data Structure:** Results are stored in a data frame with unique IDs, facilitating the calculation of a "Running Proportion" for convergence visualization.
    - *vals$history:* The data frame tracking every trial.

3. Statistical Methodology

Maximum Likelihood Estimation (MLE)

For a sequence of Bernoulli trials, the MLE for the probability of success ($\hat{p}$) is the sample proportion:

$$\hat{p}_{MLE} = \frac{\sum_{i=1}^{n} X_i}{n}; X_i = 1 \text{ if Success; otherwise, } 0$$

The application calculates this proportion dynamically, providing the most probable value for p based on the observed (sequentially simulated) data.

The Law of Large Numbers (LLN)

One of the pedagogical goals of the application is to visualize the *Weak Law of Large Numbers*. As $n \to \infty$, the frequentist estimate $\hat{p}_{MLE}$ is expected to converge to the theoretical p. The "LLN Convergence" plot tracks this sequence as the number of trials increases, supposedly highlighting the initial high variance and eventual stabilization of the proportion.

Confidence Intervals

To quantify the uncertainty of the estimate $\hat{p}$, the laboratory implements three distinct frequentist *Confidence Intervals (CI)*. Let z be the $1 - \alpha/2$ quantile of the standard normal distribution.

- Wald (Standard) Interval

$$\hat{p} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

- Wilson Score Interval

$$\frac{\hat{p} + \frac{z^2}{2n}}{1 + \frac{z^2}{n}} \pm \frac{z}{1 + \frac{z^2}{n}}\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}$$

- Agresti-Coull (AC) Interval

$$\frac{n\hat{p} + z^2/2}{n + z^2} \pm z\sqrt{\frac{\frac{n\hat{p} + z^2/2}{n + z^2}\left(1 - \frac{n\hat{p} + z^2/2}{n + z^2}\right)}{n + z^2}}$$

Comparative Theoretical Performance of the Different Confidence Intervals

Method	Robustness	Coverage Behavior
Wald	Low	Prone to "under-coverage" (too narrow) when n is small.
Wilson	High	Respects $[0,1]$ boundaries; excellent performance across all p.
Agresti-Coull	High	Conservatively stabilizes variance effectively.

4. User Guide & Operational Protocol

STEP 1: Initialize Parameters

Set your Theoretical p and Confidence Level in the sidebar.

STEP 2: Execution

Move the "Simulated Rounds" slider (e.g., to 1,000). Click "RUN SIMULATION". The values in the boxes will update immediately

STEP 3: Analyze Plots

Observe the convergence trace in the LLN plot and the interval widths in the Forest Plot.

RESET: Click "Clear Laboratory" to clear all (reactive) data before a new (generation) inquiry.

5. Technical Appendix: Programmatic Logic & Analytical Interpretation Guides

Programmatic Logic

The application uses vectorized operations for efficiency.

- **Vectorization:** Successes are calculated as *sum(outcomes)* directly from the *rbinom* vector.
- **Wilson Implementation:** The script calculates a *center* and *spread* to handle the quadratic nature of the Wilson Score to prevent the interval from exceeding the probability bounds $[0, 1]$.
- **Visualization:** *ggplot2* is utilized for all graphics. The Forest Plot employs *coord_flip()* to provide a horizontal probability scale for clearer interval comparison.

Analytical Interpretation Guidelines

When interpreting the simulation results, focus on these metrics:.

- **Coverage Success:** A "successful" interval contains the red dashed theoretical line.
- **Precision vs. Sample Size:** As n increases, the Interval Width decreases proportionally to $1/\sqrt{n}$. Thus, for instance, to halve the width, n must be quadrupled.
- **Accuracy:** The proximity of the blue MLE line to the red theoretical line.

6. Statistical and Validation Test Cases

To check the validity and stability of the application, the following test cases should be executed:.

Test Case	Scenario	Input Parameters	Expected Outcome
TC-01	Convergence Baseline	p = 0.4861, n = 5000	MLE should be within $\pm$0.02 of p.
TC-02	Confidence Expansion	Compare confidence level of 0.80 vs 0.99	Interval widths must broaden as confidence level increases.
TC-03	Boundary Stability	p = 0.01, n = 100	Wald interval may drop below 0; Wilson and AC intervals must remain > 0.
TC-04	State Persistence	Run n = 100 three times	Total trials (n) should display 300 in the purple *valueBox*.
TC-05	Reset Integrity	Click "Clear Laboratory"	All plots should clear; Total Trials should return to 0.

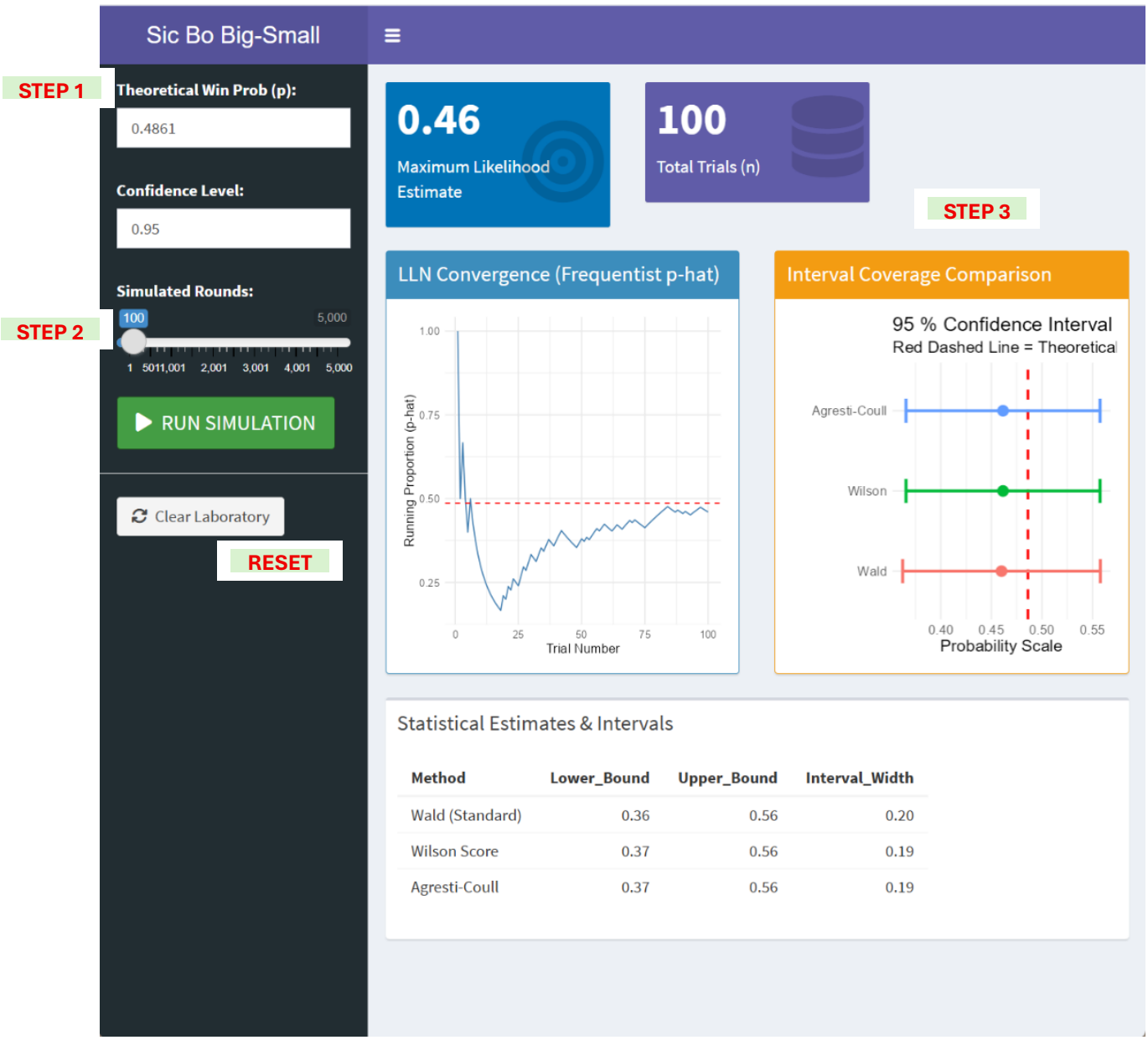

Figure 2. An annotated overview of the Sic Bo Big-Small Simulator interface highlighting the sequential pipeline, as presented in the User Guide & Operational Protocol section.